The Generalized Q-LINK Framework: Unifying Isomorphic Bases, Temporal Sparsity, and Local Observables Extension 10

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Abstract

We define the Generalized Q-LINK Framework (Extension 10) as the maximally elastic, mathematically sound expansion of the residual messenger architecture. This framework unifies an R_{yy} interaction basis, p-periodic temporal sparsity, and an m-local averaged cost operator. By evaluating the comprehensive topological, algebraic, and combinatorial properties of this union, we formally prove that these three extensions are completely mutually orthogonal. The architecture strictly guarantees an amplified asymptotic variance bound of $\mathcal{R}^{(Y)}_{p,m}(n) = \frac{n+1}{n+1-m}$, establishing a robust, hardware-agnostic hyperparameter manifold for the absolute mitigation of barren plateaus.

1 Orthogonality of the Unified Target Space

Let the Generalized Q-LINK architecture be parameterized by the tuple (Y, p, m), where Y denotes the R_{yy} collection basis, p is the layer period of the residual application, and m is the locality of the data-register observable.

We first evaluate the stability of the continuous Dynamical Lie Algebra (DLA) under the intersection of Y and p. As established, the discrete temporal sparsity p does not alter the continuous algebraic closure of the generators. Concurrently, the non-commutativity of the R_{yy} collection generators $(Y_m \otimes Y_i)$ and the distribution generators $(Z_m \otimes X_i)$ guarantees the generation of the orthogonal X_m subspace:

$$[Y_m \otimes Y_i, Z_m \otimes X_i] = 2(X_m \otimes Z_i) \tag{1}$$

Consequently, the unified DLA is strictly isomorphic to the dense R_{xx} baseline:

$$\mathfrak{g}^{(Y)}_p = \langle \mathcal{G}_{opr} \cup \mathcal{G}^{(Y)}_{coll} \cup \mathcal{G}_{dist} \rangle_{Lie} \cong \mathfrak{su}(2^{n+1}) \tag{2}$$

Because the algebraic target space remains $\mathfrak{su}(2^{n+1})$, the Haar integration volume evaluated by the Master Formula ($d = 2^{n+1}$) is mathematically invariant under both topological and temporal parameterization.

2 Invariance of the Combinatorial Shield

The traceless part of the m-local averaged cost operator, $\tilde{V}_m$, evaluates strictly on the n data qubits. Its squared Hilbert-Schmidt trace norm is a pure geometric invariant of the operator space, structurally blind to the unitary generators (Y) and temporal sequence (p) driving the state evolution:

$$Tr(\tilde{V}^2_{m,p,Y}) = \frac{2N}{\binom{n}{m}} \tag{3}$$

Because the DLA exactly saturates the identical Haar measure as the $n+1$ Vanilla baseline, the Master Formula constants cancel identically. The generalized absolute variance ratio is derived entirely from the combinatorial trace traces:

$$\mathcal{R}^{(Y)}_{p,m}(n) = \frac{Tr(\tilde{V}_m^2)}{Tr(V^2_{Vanilla,m})} = \frac{\frac{2N}{\binom{n}{m}}}{\frac{2N}{\binom{n+1}{m}}} = \frac{n+1}{n+1-m} \tag{4}$$

Theorem of the Grand Unified Survivor: The trace combinatorial shield $\frac{n+1}{n+1-m}$ is mathematically immune to simultaneous basis isomorphism and temporal sparsity. The three extensions are strictly algebraically orthogonal.

3 The Physical Hyperparameter Manifold

Equation (4) proves that the Generalized Q-LINK Framework avoids the catastrophic $O(2^{-k})$ Haar volume penalty of multi-messenger expansion, while providing experimentalists with three independent axes of control:

- **Topological Adaptation (Y):** The R_{yy} basis allows the architecture to be mapped natively onto specific physical hardware (e.g., YY-coupled superconducting circuits) without fracturing the residual channel's Lie algebra.
- **Dynamical Throttling (p):** The period p explicitly controls the rate of operator spreading. By fractionating the causal lightcone, the depth required to reach Haar saturation scales as $D_{sat} \propto p \cdot \mathcal{O}(poly(n))$, allowing the optimizer a wider, shallower trajectory window before convergence.
- **Geometric Floor (m):** The locality m strictly governs the asymptotic boundary. By increasing m, the combinatorial trace advantage actively shields the gradient magnitude from the $(n+1)$ measurement dilution experienced by Vanilla circuits.

4 Conclusion

The Q-LINK topology is fundamentally brittle to structural additions (multi-messengers) or basis ablations (R_{zz}), which induce fatal algebraic collapse. However, it is profoundly elastic to spatial, temporal, and isomorphic parameterization. The Generalized Framework presented here represents the maximally viable theoretical extension of the residual messenger architecture, offering a mathematically guaranteed, fully tunable blueprint for finite-depth VQA trainability.